Span and independence

1 Linear combinations and span

Definition 2.1 Linear combination and span
A linear combination of $v_1, \dots, v_m$ is a vector $a_1 v_1 + \cdots + a_m v_m$, where the coefficients belong to the scalar field $\mathbb{F}$. The span of the list is the set of all these combinations. Throughout these notes, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.

$$\text{span}(v_1, \dots, v_m) = \{a_1 v_1 + \cdots + a_m v_m : a_1, \dots, a_m \in \mathbb{F}\}$$

The list spans V if its span equals V. This means every vector of V has at least one representation using the list

Example 2.2 A plane described in two ways

Let $u = (1,0,1)$ and $v = (0,1,1)$. A general linear combination is

$$au + bv = (a, b, a + b)$$

Thus, every generated vector satisfies $z = x + y$. so $\text{span}(u, v) = \{(x, y, z) \in \mathbb{R}^3 : z = x + y\}$.

Remarks. The span of the empty list is $\{0\}$. A finite-dimensional vector space is one that has a finite spanning list.

2 Why span is a subspace

Theorem 2.3 The smallest containing subspace
The span of a list is the smallest subspace containing every vector in the list.

Proof. Let $S = \text{span}(v_1, \dots, v_m)$. Choosing all coefficients zero shows that 0 belongs to S. If x and y belong to S, write

$$x = a_1 v_1 + \cdots + a_m v_m, \qquad y = b_1 v_1 + \cdots + b_m v_m$$

For arbitrary scalars α and β, regrouping gives

$$\alpha x + \beta y = (\alpha a_1 + \beta b_1) v_1 + \cdots + (\alpha a_m + \beta b_m) v_m$$

This is another linear combination of the same list. The subspace criterion from Week 1 therefore applies. Each v_j belongs to S by choosing its coefficient to be 1 and all the others to be 0.

Now suppose U is a subspace containing every v_j. Closure under scaling and addition puts every linear combination in U. Hence $S \subseteq U$, which proves the minimality claim.

3 Linear independence and uniqueness

Definition 2.5 Linear independence
A list $v_1, \dots, v_m$ is linearly independent if $a_1 v_1 + \cdots + a_m v_m = 0$ forces $a_1 = \cdots = a_m = 0$. It is dependent if a zero relation exists with at least one nonzero coefficient.

Theorem 2.6 Unique coefficients
A list is linearly independent if and only if every vector in its span has a unique representation as a linear combination of the list.

Proof. Suppose the list is independent and a vector has two representations. Subtract them:

$$(a_1 - b_1) v_1 + \cdots + (a_m - b_m) v_m = 0$$

Independence forces $a_j = b_j$ for every j. Conversely, if representations are unique, zero can only have the representation with all coefficients zero. This is exactly independence.

Example 2.7

In $\mathbb{R}^2$, the vectors $(1,0)$, $(0,1)$ and $(1,1)$ are dependence. Since,

$$(1,0) + (0,1) - (1,1) = (0,0)$$

A list containing zero is dependent: put coefficient 1 on zero and 0 on the other entries. Every sublist of an independent list is independent, since any relation on the sublist extends to the original list by adding zero coefficients.

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Lemma 2.9 Deleting a redundant vector
If $a_1v_1 + \cdots + a_mv_m = 0$ and $a_j \neq 0$, then deleting v_j does not change the span of the list.

Proof. Solve the relation for v_j. It is a linear combination of the other vectors. In any combination of the full list, substitute this expression wherever v_j occurs. The result uses only the remaining vectors. The reverse containment is automatic because the shorter list is part of the original list.

The ordered form of the dependence lemma

Choose the largest index k whose coefficient in a nontrivial zero relation is nonzero. All later coefficients vanish, and rearrangement gives

$$v_k = -\frac{a_1}{a_k}v_1 - \cdots - \frac{a_{k-1}}{a_k}v_{k-1}$$

Thus some vector lies in the span of its predecessors and can be removed. If $k = 1$, this says $v_1 = 0$, which belongs to the span of the empty list.

Exercises

1. Consider $(1,1,0)$, $(0,1,1)$, $(1,2,1)$. Decide whether the list is independent and describe its span by one equation.

2. Let $p_1 = 1 + x$ and $p_2 = x + x^2$. Decide whether $2 + 5x + 3x^2$ and $1 + x^2$ belong to $\text{span}(p_1, p_2)$.

3. Prove that every list of four vectors in $\mathbb{R}^3$ is dependent.